

Creating Heat and Light

A Canadian program tests the efficiency and future potential of residential combined heat and power systems at its residential building test center.

**by Mike Bell
and John Gusdorf**

Combined heat and power systems (CHP), which jointly produce heat and electricity from a single fuel source, can be up to 90% efficient, compared with conventional U.S. power production, which converts only 35% of fuel energy into delivered electricity. The rest of the fuel energy is normally lost in the form of heat. By using this heat to produce space heat and hot water, CHPs can decrease energy costs and increase fuel use efficiency.

Currently, CHP units in the United States are used almost exclusively for commercial and industrial buildings. However, there is tremendous potential for using this technology in residences as well. Recently, the Department of Energy (DOE) assembled a team of appliance industry leaders to develop a CHP system to provide more reliable and less costly energy for residential use. Projects like this show that residential CHP systems are a likely reality in the future of home heating and power.

Advancing the Home CHP Cause

CHP isn't just a future reality—residential systems already exist. One such system for individual residences was recently tested through a joint effort of Natural Resources Canada, National Research Council Canada, and Canada Mortgage and Housing Corporation at



The Stirling engine (shown here with Mike Bell and with the cover off), an external-combustion device capable of burning multiple fuels, was chosen for a residential CHP system.

cently accelerate development and residential integration of these systems (see "Canadian Centre for Housing Technology," p. 37).

The CCHT offers an intensively monitored real-world environment, with simulated occupancy, to assess the performance of residential CHP systems in secure premises. Simulated occupancy included such measures as turning lights, showers, and appliances on and off daily. Separate lights in several locations were also turned on to simulate heat from occupants.

Residential CHP systems that are currently available or are being developed are based on internal combustion engines (ICE), on Stirling engines, or on fuel cells. ICE is a proven technology, but it creates noise, odors, and pollution. Fuel cells hold great promise, but they are still in the prototype stage. For this project, we integrated into a residential building and tested a near-commercially available, natural gas fired, Stirling engine CHP that is sized for individual residences.

Home CHP Basics

A residential CHP system must be integrated into the house's electrical, space-heating, and domestic hot water (DHW) systems. The most common space heating system in homes is a natural gas furnace with forced air. For the CHP project, an air handler (fan plus heat exchanger) was used to supply heated air through the same ductwork.

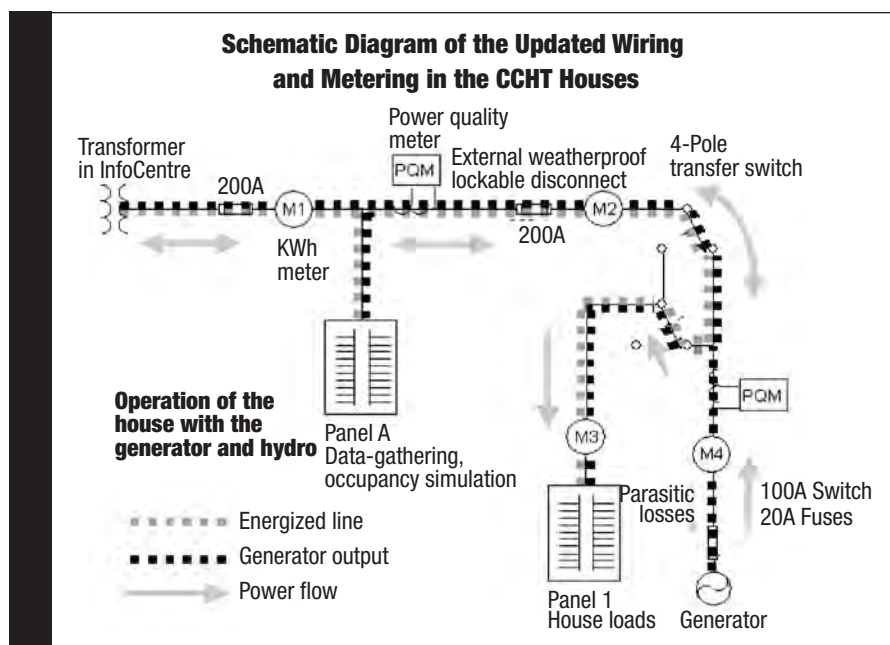


Figure 1. Electrical modifications integrating the CHP system into the house's electrical system were required to allow the CHP to export electricity to the grid.

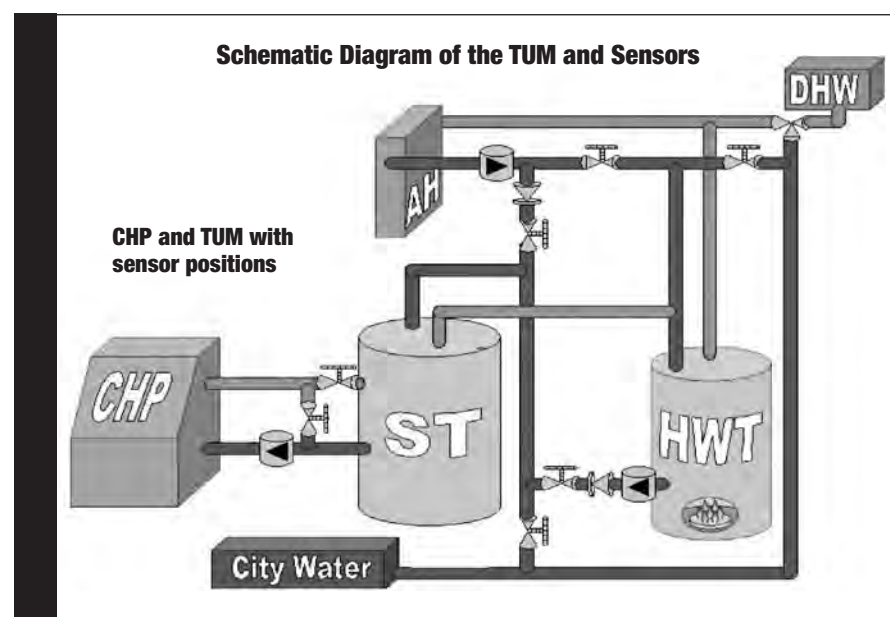


Figure 2. The thermal utilization module (TUM) consists of the storage tank (ST), the hot water tank (HWT), the air handler (AH), and pumps and a mixing valve. The burner on the HWT served as a backup or top-up burner in instances where the CHP unit could not supply all of the heat requirements, due either to heavy demand or to shut-down of the CHP unit. The TUM was designed so that it could be operated in two different setups by opening or closing the four valves between the ST and the HWT. This made it possible to determine whether the strategy for directing heated water and cooler return water affected the operating efficiency of the CHP and the TUM.

Other issues that were raised included sizing the CHP system relative to average and peak electrical and thermal loads; deciding whether the CHP system

is controlled by demands for electricity (electrical lead) or heat (thermal lead); dealing with the simultaneous production of heat and electricity; opti-

mizing CHP system operation and run time; determining the need for, and the sizing of, thermal and electrical storage (especially for off-grid systems); dealing with the handling and use of possible excess heat in summer; and specifying grid connection techniques.

The primary objective of our project was to develop and demonstrate a test facility at the CCHT that could assess residential CHP systems, and their integration into houses, under real-world conditions. A secondary objective was to assess the performance of various residential CHP systems under real-world conditions.

To integrate the CHP system into the house's various existing systems, we made numerous modifications to accommodate residential CHP systems. These included electrical modifications to integrate the CHP system into the house's electrical system and to allow the CHP to export electricity to the grid; the design and installation of a thermal utilization module (TUM) to integrate the CHP system into the house's space- and water-heating systems; the design and installation of a combined monitoring and control system for the CHP and TUM; and the installation and connection of the CHP unit. The electrical modifications accommodated the installation of CHP systems with a generating capacity of up to 40 kW for either grid-dependent operation or stand-alone, grid-independent operation. Although only one of the two CCHT houses was used in this project, the electrical modifications were made to both of them, so that in the future, two CHP units can be tested simultaneously.

The cost of these wiring modifications was \$5,068 (CAN \$6,450) for each house. Because these houses were modified to allow for various types of CHP system, both connected and off-grid, and sized up to 40 kW, the wiring scheme is more elaborate than would be the case for a normal residential installation. It is estimated that the cost to retrofit a typical residence would range from \$1,500 to \$2,500 (CAN\$2,000 to CAN\$3,000). This cost could be lower if simpler installations, such as the elimination of the external disconnect switch, were acceptable to electrical authorities.

The Stirling Engine Combined Heat and Power System

Natural Resources Canada purchased the natural gas-fired Stirling engine before the start of this project for \$12,000 (current models are \$7,500). A Stirling engine is an external-combustion device, which can burn many different fuels—diesel, natural gas, propane, biogas, kerosene, and solid fuels—provided that the heat exchanger is specifically designed for the selected fuel. It has an induction motor/generator that automatically synchronizes the frequency of its alternating-current output to the grid, and cannot operate unless it is connected to an active grid. The Stirling CHP unit is a heat-led device; it is turned on when there is a need for heat in the TUM.

The TUM controls turn the circulation pump on and off according to the temperatures in the storage tank and the hot water tank. The pump is turned on if the hot water tank needs heat and can get it from the storage tank, and if the bottom of the hot water tank is between 116° and 158°F. The house thermostat controls the operation of the air handler pump and fan, and the demand for hot water is controlled by the house's simulated-occupancy system.

Each heat transfer loop of the TUM has two thermocouples and a flowmeter. As shown in Figure 2, the loops run from the CHP unit to the storage tank; from the storage tank to the hot water tank; from the hot water tank to the air handler; and from the hot water tank to hot water taps. Data were collected, and heat flows were calculated, every ten seconds. These heat flows were averaged and saved every minute. Two existing natural gas meters were used to monitor the gas consumption of the CHP unit and that of the hot water tank backup burner.

Reviewing the Data

Monitoring took place between March 13 and June 10, 2003. During the review and analysis of results, we realized that the amount of heat required by the house for space and water heating would affect each component of the CHP system. Accordingly,

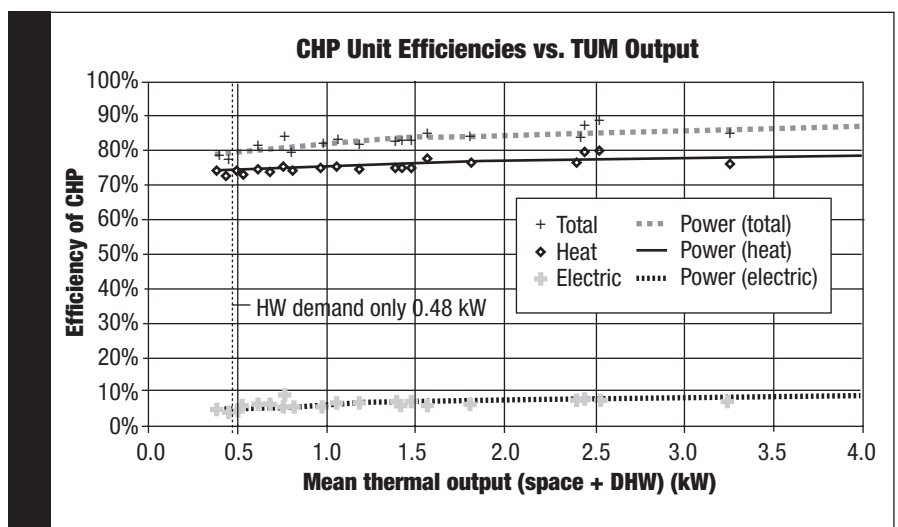


Figure 3. The efficiency of the CHP is nearly constant, increasing only slightly with output from the TUM.

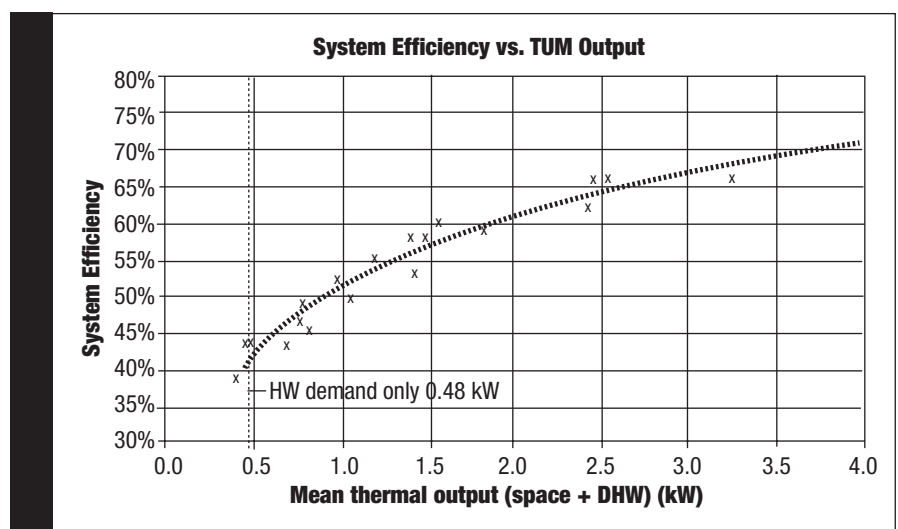


Figure 4. System efficiency increases significantly with TUM output. Heat losses are fairly constant, and at low outputs they are a larger percentage of the output.

each of the efficiencies was plotted against the thermal output of the TUM (see Figures 3 and 4). The performance of the CHP unit and the performance of the TUM both depend, to varying degrees, on the TUM output. The same daily pattern of hot water demand (260 liters per day) was used for all 27 runs monitored. When there was no demand for space heating, (hot water demand only), the average demand on the TUM is 0.48 kW. For the ten runs with no space heating, thermal utilization module output varied from 0.39 to 0.62 kW. This variation was due to different start

and end times caused by hot water demands, and to different CHP unit run time patterns. All curves in Figures 3 and 4 are projected to reach close to the TUM output capacity of 4.40 kW.

Emerging Trends

CHP unit efficiency increased only slightly with system output. The two factors that could affect unit efficiency are the temperature of the inlet water and the length of time the unit runs. Inlet water should have been cooler, on average, when demand for space heating and



The CCHT includes two thermally identical R2000 houses with simulated occupancy and intensive monitoring, plus a three unit townhouse.

Table 1. CHP Unit Efficiencies (%)

	Heat	Electric	Total
Minimum	73.1	4.0	77.5
Mean	75.5	6.4	81.7
Maximum	80.5	9.0	88.7
Mean, Setup 1	76.2	6.1	81.9
Mean, Setup 2	75.1	6.6	81.6

hot water was higher, but we did not analyze inlet water temperature, and apparently the unit is not very sensitive to it. Internal temperatures are around 750°F, so a difference of a few tens of degrees in the inlet should not make much difference.

Table 2. TUM and Total System Efficiencies (%)

	TUM	System
Minimum	44.2	38.9
Mean	56.8	49.8
Maximum	75.1	66.4
Mean, Setup 1	57.4	50.3
Mean, Setup 2	56.4	49.5

As the length of time that the unit runs before shutting off decreases, the efficiency of the unit should drop, due to the use of electricity and natural gas in start-up and of electricity in shut-down. But even with no space heating, the runs were at least 70 minutes long,

so the efficiency drop was not significant. It might be greater in a unit with a smaller storage capacity.

CHP unit heating efficiency varied from 74% for hot water only to about 79% at system capacity. CHP unit electrical efficiency varied from 5.5% to about 9% at system capacity, and the unit's total efficiency varied from 79.5% to about 88% at system capacity (see Figure 3).

System energy efficiency depends significantly on thermal load (see Figure 4). This is because the standby losses from the TUM are relatively constant, while the useful heat varies mainly with space heat load. System efficiency varies from 41% for hot water only to over 70% at CHP capacity. System efficiency compares favorably with the efficiency (energy factor) of domestic water heaters, while generating by-product electricity. These efficiencies could be improved by improving the design of the TUM. As shown in Tables 1 and 2, there were only small differences between the two setups investigated, although setup 2 appeared to generate a slightly better electrical efficiency than setup 1.

Improvements to the TUM efficiency could be based on modeling to determine whether two tanks are needed, or whether one larger one would do. If two are needed for some applications, one tank could be left unheated during seasons when no heating is needed, thus reducing heat losses

Canadian Centre for Housing Technology

Canada is fortunate in that its government operates a unique residential test facility known as the Canadian Centre for Housing Technology (CCHT). This facility, located in Ottawa, consists of two thermally identical R2000 houses (houses built to be 40% more efficient than Canadian building code requirements) and a three-unit town house. The test houses are highly instrumented, with approximately 230 points where temperatures, relative humidity, air flows, and moisture content are measured every five

minutes. There are also 28 kWh meters, 10 water meters, and four natural gas meters that save data every five minutes. The houses have programmable simulated occupancy which controls loads such as lights, stove, clothes washer and dryer, and dishwasher, as well as hot and cold water draws. The heat produced by the occupants is simulated by banks of light bulbs operated in different parts of the house according to the time of day.

The CCHT provides an ideal stepping-stone between lab testing of residential

CHP systems and field trials in occupied housing, as it provides a real-house installation and energy demand operating in real weather; realistic simulated occupancy, which is repeatable; intensive monitoring; easy access to equipment; and a secure site. Part of our program has been to adapt the CCHT to accept CHP systems for quick installation, since appropriate buildings and monitoring systems are already in place. This facility is unique to Canada and is available for use by manufacturers, utilities, and industry.

by approximately one-half. Similarly, a single stratified tank could have only its top half heated during warm seasons. Control of the hot water tank burner should be integrated into the TUM control, rather than being left to the aquastat on the original tank.

The TUM includes three pumps that consume between 73 and 86 watts each. The CHP pump uses an average of 1.19 kWh per run, or 7.8% of the CHP net electrical output, per run. All three pumps together average 1.95 kWh, per run, which is 12.4% of the CHP net output, or 0.4% of the system output (heat plus net CHP electric). The pump energy is therefore low compared to CHP electrical generation, and insignificant compared to total output. Furthermore, pump energy could be reduced if the pumps were optimized in terms of size, efficiency, and control, and would be even lower if the electrical output of the CHP were higher, as it would be with a fuel cell sized for a house.

Two examples were selected to highlight how much electricity the CHP generates relative to house electricity demand, and to show where that electricity goes. The first example represents one of the colder periods of testing, while the second run represents a milder day (see Figure 5 and Table 3). Figure 5 shows house demand, supply by the CHP unit, and net house demand from the grid. The reduction in electrical requirements from the grid due to the CHP unit is apparent, as are periods of electricity being exported to the grid (negative grid supply). Table 3 summarizes the electricity balance for these two runs. While most of the electricity produced by the CHP unit goes to the house (94% and 98% in these examples), there are still instances of export to the grid (6% and 2%), even with this small heat-led generator. The CHP unit supplies important percentages of the house's electricity requirement (43% and 25% in these examples).

Table 3. Example Electricity Balance for the CHP Supply and the House Demand

	Run 13 (April 1 & 2)		Run 23 (April 24)	
	kWh	%	kWh	%
CHP Unit				
CHP electricity generation	9.95	100	5.34	100
CHP electricity used by the house	9.36	94	5.23	98
CHP electricity exported to the grid	0.60	6	0.11	2
House				
Total house electricity consumption	21.74	100	20.72	100
Electricity supplied by the grid	12.38	57	15.49	75
Electricity supplied by the CHP	9.36	43	5.23	25

**Electrical Power: House Demand, CHP Supply, and Grid Supply
Run 13, Setup 1, April 1 and 2, 2003**

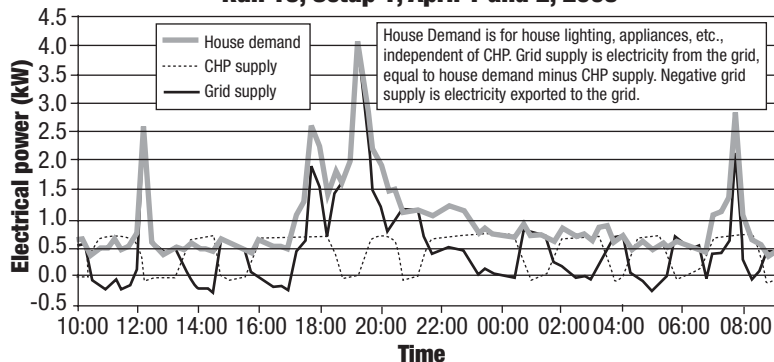


Figure 5. This graph shows the electrical power profile for the CHP and TUM, run 13.

The Future of CHP Use

We believe that this technology has a high potential for use both in new houses and as a retrofit in existing houses. That's why we will soon be starting a demonstration project that will test the ability of a newer version CHP to run unattended for a year. The project will monitor the new unit's gas use and its heat and electrical output. The high initial cost for homeowners may require utility sponsorship, as is happening in the United Kingdom. Utilities should be interested, because having large numbers of CHP units in their area would reduce their peak demands, at least in the winter. Restaurants and cafeteria, with their large, more or less continuous, and year-

round demand for hot water for dishwashing, represent another potential market for CHPs.



Mike Bell provides leadership to the Buildings Group of Natural Resources Canada in the integration and performance of combined heat and power systems into residential buildings. He also develops other strategic projects in areas such as novel PV systems for houses.

John Gusdorf is a research specialist with the Buildings Group of Natural Resources Canada. He works in data collection and analysis related to energy efficiency and indoor air quality in Canadian housing. His recent work has included testing and analysis of energy-efficient furnace fan motors and combined heat and power systems for residences.